🤖 AI Olympiad Book – Std 4

Units: AI Basics • Where Is AI Hiding? • Meet the Smart Machines

25 pages • Fun facts • Activities • Practice quiz

Page 1 – Welcome, Young Scientist! 🚀

🖼️ **Image idea:** A smiling blue robot waving from a rocket, with stars, a laptop and a trophy around it.

Hello, champion! Get ready for an amazing journey into the world of **Artificial Intelligence (AI)**.

By the end of this book you will be able to:

- Explain what AI is in your own words
- Spot AI hiding in everyday places
- Meet smart machines and understand how they work
- Win your AI Olympiad! 🏆

Page 2 – Meet Your Guide: Robo-Pi 🤖

🖼️ **Image idea:** A cute round robot with big eyes, a graduation cap and a magnifying glass.

Hi! I am **Robo-Pi**, your friendly guide. Look for my signs in every chapter:

- 💡 **Did You Know?** – a surprising fun fact
- 🧠 **Think Fast!** – a quick question
- ✏️ **Try It!** – an activity
- ⭐ **Remember** – the big idea of the page

PART 1: AI BASICS 🧠

Page 3 – What Is Intelligence?

🖼️ **Image idea:** A child solving a puzzle, a dog fetching a ball, a bird opening a nut.

Intelligence means being able to **learn, think, understand and solve problems**.

Humans are intelligent. You learn new words, solve maths sums and make friends. Animals are smart too: crows can use sticks as tools, and dogs can learn many commands.

🧠 **Think Fast!** Name 3 things you learned this week.

⭐ **Remember:** Intelligence = learning + thinking + solving problems.

Page 4 – So, What Is AI?

🖼️ **Image idea:** A computer with a glowing brain inside, connected by lines to a phone, a car and a robot.

AI stands for Artificial Intelligence.

- *Artificial* = made by people, not born in nature
- *Intelligence* = the ability to learn and solve problems

So AI is a **computer or machine that can do smart things**, like recognising pictures, understanding speech or making decisions.

💡 **Did You Know?** The name "Artificial Intelligence" was first used by scientists in 1956, a long time before you were born!

⭐ **Remember:** AI = machines that can learn and act smart.

Page 5 – Human Brain vs Machine Brain

🖼️ **Image idea:** Split picture. Left: a child's head with a brain. Right: a robot head with circuits.

	🧒 Human Brain	🤖 AI
Learns from	Experience, school, play	Lots of examples (data)
Feelings	Yes! Happy, sad, brave	No feelings
Creativity	Imagines new ideas	Mixes patterns it has seen
Tiredness	Needs sleep	Never gets sleepy
Speed with huge data	Slow	Very fast

⭐ **Remember:** AI is very fast at some jobs, but humans have feelings, kindness and imagination.

Page 6 – Data: The Food of AI 🍎

🖼️ **Image idea:** A hungry robot eating pictures, numbers and words from a big plate.

You need food to grow. AI needs **data** to learn.

Data means information such as photos, numbers, words, sounds and videos.

- More good examples → smarter AI
- Wrong or unfair examples → confused AI

💡 **Did You Know?** To learn what a cat looks like, an AI may look at thousands of cat photos!

✏️ **Try It!** List 3 kinds of data: one you can see, one you can hear, one you can count.

Page 7 – How Does a Machine Learn? 🐱🐶

🖼️ **Image idea:** Three steps drawn as cartoon panels: photos of cats and dogs → robot studying → robot correctly saying "Cat!"

Let's teach a computer to tell cats from dogs:

1. **Show examples** – photos labelled "cat" or "dog"
2. **Practise** – the computer finds patterns (pointy ears, whiskers, snouts)
3. **Test** – we show a new photo and it guesses
4. **Improve** – if it is wrong, it learns and tries again

This way of learning from examples is called **Machine Learning**.

🧠 **Think Fast!** How is this like you learning to recognise fruits?

Page 8 – Algorithm: A Recipe for Computers 📜

🖼️ **Image idea:** A recipe card with numbered steps next to a robot chef making a sandwich.

An **algorithm** is a list of **step-by-step instructions** to finish a task.

Algorithm to make a sandwich:

1. Take two bread slices
2. Spread butter
3. Add vegetables
4. Close the sandwich

5. Enjoy! 😊

If the steps are in the wrong order, the result is a mess!

✏️ **Try It!** Write an algorithm for brushing your teeth in 5 steps.

⭐ **Remember:** Computers follow algorithms. The order of steps matters.

Page 9 – Oops! AI Can Make Mistakes ⚠️

🖼️ **Image idea:** A robot confused, holding a muffin while a picture of a puppy is beside it with a question mark.

AI is clever, but it is **not perfect**.

- It can guess wrong if it did not see enough examples
- It can be unfair if its data was unfair
- It does not truly "understand" like a person

That is why humans must always **check** what AI says and decide wisely.

💡 **Did You Know?** Some AI programs once mixed up photos of chihuahuas and blueberry muffins because they look a bit alike!

⭐ **Remember:** Use AI as a helper, but think for yourself.

Page 10 – Part 1 Brain Teasers 🧩

🖼️ **Image idea:** A colourful maze with a robot at the start and a trophy at the end.

A. Fill in the blanks

1. AI stands for _____ Intelligence.
2. Information that AI learns from is called _____.
3. Step-by-step instructions are called an _____.

B. True or False

1. AI has feelings like humans. ( )
2. AI can make mistakes. ( )
3. More good data helps AI learn better. ( )

C. Odd one out: Photo, Sound, Pencil, Video

(Answers on Page 25)

PART 2: WHERE IS AI HIDING? 🔍

Page 11 – Let's Go on an AI Hunt! 🕵️

🖼️ **Image idea:** A detective robot with a magnifying glass looking at a busy street full of hidden glowing symbols.

AI is like a **secret helper**. It hides in many things you use every day, and you may not even notice it!

In this part we will become **AI detectives**.

🧠 **Think Fast!** Do you think a torch is AI? Why or why not? (Hint: does it learn or decide?)

Page 12 – AI in Your Phone 📱

🖼️ **Image idea:** A big smartphone with icons popping out: camera, keyboard, voice mic, photos.

Your phone has lots of AI:

- **Face unlock** – recognises your face
- **Camera** – makes photos brighter and sharper
- **Keyboard** – suggests the next word
- **Voice assistant** – listens and answers
- **Photo gallery** – groups photos by people or places

⭐ **Remember:** Your phone uses AI to understand pictures, words and voices.

Page 13 – AI in Videos and Games 🎮

🖼️ **Image idea:** A child watching a tablet, with a robot hand picking video thumbnails from a shelf.

- **Video apps** suggest videos you may like, based on what you watched before. This is a **recommendation system**.
- **Games** use AI to control computer players, so enemies seem to think and move cleverly.
- **Filters** on cameras find your face and add fun stickers.

💡 **Did You Know?** The "recommended for you" row is not chosen by a person. A computer program picks it!

Page 14 – AI in Maps and Travel 🗺️

🖼️ **Image idea:** A map with a glowing route, traffic cars in red and green, and a small robot pointing the way.

Map apps use AI to:

- Find the **fastest route**
- Guess how heavy the **traffic** is
- Tell you the **time** you will reach

It does this by studying lots of data from many roads and travellers.

✏️ **Try It!** Ask an adult to show you a map app. Can you spot the fastest route?

Page 15 – AI at Home 🏠

🖼️ **Image idea:** A cutaway house showing a smart speaker, robot vacuum, smart TV and washing machine, each with a tiny glowing brain.

Smart helpers at home:

- 🔊 **Smart speaker** – plays songs when you ask
- 🧹 **Robot vacuum** – maps the room and avoids furniture
- 📺 **Smart TV** – recommends shows and understands voice
- 🌡️ **Smart AC / fridge** – adjusts settings to save energy

⭐ **Remember:** "Smart" gadgets usually have some AI inside.

Page 16 – AI in Hospitals and Farms 🏥🌾

🖼️ **Image idea:** Left: a doctor looking at a scan with AI highlights. Right: a drone flying over green fields.

In hospitals: AI helps doctors look at X-rays and scans to spot problems early. The doctor still makes the final decision.

On farms: Drones and cameras with AI check if crops are healthy, if plants need water, and where pests are hiding.

💡 **Did You Know?** AI helps farmers grow more food using less water!

Page 17 – AI at School and Shopping 🏫🛒

🖼️ **Image idea:** A classroom with a learning app on a tablet, and a shopping website showing suggested items.

- **Learning apps** notice which questions you find hard and give you more practice.
- **Translation apps** change words from one language to another.
- **Online shops** suggest things similar to what you looked at.
- **Spell-check** underlines mistakes in your writing.

🧠 **Think Fast!** Which of these have you used?

Page 18 – Part 2 Activity: AI Detective 🔎

🖼️ **Image idea:** A checklist page with magnifying glass stickers and a robot stamp saying "AI FOUND!"

✏️ **Try It!** Tick (✔) the things that use AI. Cross (✘) the ones that don't.

Item	AI?
Face unlock on a phone	
A wooden ruler	
Video app suggestions	
A simple pencil	
Map app showing traffic	
A ceiling fan with only a switch	

Bonus: Draw one place in your home where AI might be hiding!

PART 3: MEET THE SMART MACHINES 🦾

Page 19 – What Is a Robot? 🤖

🖼️ **Image idea:** Different robots standing in a line: arm robot, wheeled robot, humanoid robot, tiny toy robot.

A **robot** is a machine that can **do jobs by itself or with a little help from humans**.

Not every robot looks like a person. Many robots are arms, wheels or flying machines!

💡 **Did You Know?** Many robots work in factories, building cars and gadgets all day.

⭐ **Remember:** A robot can have AI, but not every robot is very smart. Some only repeat the same steps.

Page 20 – Sense, Think, Act! 👀🧠💪

🖼️ **Image idea:** A robot with arrows pointing to its eyes (camera), brain (computer) and arms (motors).

Smart machines work in **3 steps**:

1. **SENSE** 👀 – cameras, microphones and sensors collect information
2. **THINK** 🧠 – the computer (with AI) decides what to do
3. **ACT** 💪 – motors, wheels or speakers carry out the action

Example – Robot vacuum: Sense: finds a wall → Think: "turn left" → Act: turns and keeps cleaning.

✏️ **Try It!** Write the Sense-Think-Act steps for a **traffic light that detects cars**.

Page 21 – Self-Driving Cars and Drones 🚗🚁

🖼️ **Image idea:** A futuristic car with sensor waves around it and a drone carrying a small parcel.

Self-driving cars use cameras and sensors to see roads, signals and people, then decide when to stop, go or turn. Humans still work hard to make them safe.

Drones are flying machines. Some are used to take photos, check farms or deliver light packages.

🧠 **Think Fast!** Which senses (eyes, ears) does a self-driving car need most?

Page 22 – Helper Robots 🧹🏭🔭

🖼️ **Image idea:** Three scenes: a robot vacuum at home, factory robot arms, a space rover on a red planet.

- 🧹 **Cleaning robots** tidy floors in homes and offices.
- 🏭 **Factory robots** lift heavy things and do repeated work fast.

- 🔭 **Space rovers** explore planets like Mars where humans cannot easily go.
- 🌊 **Underwater robots** explore deep oceans.

💡 **Did You Know?** Robots are great for jobs that are dangerous, dull or very tiny!

Page 23 – Chatbots and Voice Assistants 💬🎙️

🖼️ **Image idea:** A child talking to a speech bubble robot, with question and answer bubbles floating.

Chatbots are AI programs that **talk or type with you**. **Voice assistants** listen to your voice and answer.

They can:

- Answer questions
- Help with homework ideas
- Tell stories and jokes
- Translate words

⚠️ **But remember:** A chatbot can be wrong. Always check important answers with a book, teacher or parent.

✏️ **Try It!** With an adult, ask a voice assistant a fun question. Was the answer right?

Page 24 – Using AI Safely and Kindly 🛡️💛

🖼️ **Image idea:** A child and a robot shaking hands under a shield with a heart on it.

Golden rules for smart kids:

1. 🔒 **Never share** private information (address, passwords, school name) with AI.
2. 👨‍👩‍👧 **Ask an adult** before using new apps.
3. ✅ **Check facts.** AI can be wrong.
4. 🧠 **Do your own thinking.** Use AI to help, not to copy.
5. 💛 **Be kind.** Use polite words, and never use AI to hurt anyone.

⭐ **Remember:** Humans are the bosses of machines!

Page 25 – AI Olympiad Practice Test 🏆

🖼️ **Image idea:** A golden trophy, medals and a robot holding a "Well Done!" sign.

Choose the correct answer.

1. AI stands for: a) Active Internet b) Artificial Intelligence c) Automatic Image d) Amazing Idea
2. Information that helps AI learn is called: a) Data b) Dust c) Dream d) Dance
3. A list of steps to do a task is a(n): a) Picture b) Algorithm c) Battery d) Robot
4. Which of these uses AI? a) Wooden ruler b) Plain notebook c) Face unlock on phone d) Chalk
5. The 3 steps of a smart machine are: a) Eat, Sleep, Play b) Sense, Think, Act c) Read, Write, Draw d) Run, Jump, Hop
6. Which can explore Mars? a) Space rover b) Toy car c) Bicycle d) Kite
7. If a chatbot gives an important answer, you should: a) Believe it blindly b) Check it c) Ignore all answers d) Share your password
8. Video apps suggesting videos use a: a) Recommendation system b) Pencil box c) Paint brush d) Compass

✅ Answer Key

Practice test: 1-b, 2-a, 3-b, 4-c, 5-b, 6-a, 7-b, 8-a

Page 10 Brain Teasers: A. 1) Artificial 2) Data 3) Algorithm. B. 1) False 2) True 3) True. C. Pencil (it is not data).

Page 18 AI Detective: ✔ Face unlock, video suggestions, map traffic. ✘ Wooden ruler, pencil, simple ceiling fan.

🎉 **Congratulations, AI Champion! You are ready for the Olympiad!** 🎉